How-To Library

Informatica

Informatica Port Administration

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Abstract

Domain and application services ports can either be static ports or dynamic ports. The Informatica domain and domain components are assigned to static ports. Certain application services are also assigned to static ports while others run on dynamic ports.

Supported Versions

- Informatica Data Quality 10.0
- Informatica Data Services 10.0
- Big Data Management 10.0
- PowerCenter 10.0

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Informatica Port Administration Overview

When you install the Informatica domain and server, the installer sets up the ports for components in the Informatica domain. During the same installation, the installer designates the range of ports to use for application service processes that run on the node where you install Informatica.

Static and Dynamic Ports

Domain and application services ports can either be static ports or dynamic ports.

Static ports are ports that are assigned dedicated port numbers that do not change. The Informatica domain and domain components are assigned to static ports. Certain Informatica services are also assigned to static ports; these services will always start on the static ports.

Dynamic ports are ports that are not assigned port numbers. Instead, the Informatica Service Manager dynamically allocates port numbers to each of the application services processes as they start up.

Dynamic Port Behavior

Certain Informatica application services use port numbers that are dynamically allocated upon service startup.

When the application services and processes start up, the Informatica Service Manager assigns port numbers from a range of pre-defined port numbers to the processes. The Service Manager starts with the minimum port number of the pre-defined range and incrementally works its way up the range to find the first available port. The Service Manager repeats this process for each of the application services as they start up.

You define the range of port numbers in the domain when you install Informatica. The port range starts with the minimum port number and ends with the maximum port number.

Default Number of dynamic ports	100
Default Minimum Port Number	6014
Default Maximum Port Number	6114

Dynamic Port Design

The Informatica mechanism of dynamic port allocation across pre-defined ranges has been carefully designed over a multi-year period to handle a wide range of scenarios. There are several benefits to this design:

- Dynamically assigning the port number upon service startup means that users do not have to manually assign port numbers to the services they create.
- Setting the range of port numbers gives users the flexibility to set the range to a desirable size.
- Limiting the dynamic ports to a specific range of numbers allows the IT team to efficiently troubleshoot issues.

Informatica Domain Ports

The Informatica domain and domain components are assigned to static ports. The following table lists the default port numbers for each of the domain components:

Port Name	Default Port Number(s)
Node Port	6005
Service Manager Port	6006
Service Manager Shutdown Port	6007

Port Name	Default Port Number(s)
Informatica Administrator Port	HTTP: 6008 HTTPS: 8443
Informatica Administrator Shutdown Port	6009

If you do not manually assign the port numbers during the installation, Informatica uses the default values.

Informatica Services Ports

The Informatica services run on static or dynamic ports. For application services that run on static ports, you manually assign the port number when you create the application service. If a service does not start on a static port, the Informatica Service Manager dynamically assigns a port number to the service.

The following table lists the Informatica services and the static and dynamic port details:

Service Name	Static or Dynamic?	Port Numbers
Analyst Service	Static	HTTP: 8085
Content Management Service	Static	HTTP: 8105
Data Integration Service	Static	HTTP: 8095 HTTPS: No default port number. Manually assigned during service creation.
Metadata Manager Service	Static	MM: 10250 MM Agent: 10251
Model Repository Service	Dynamic	Dynamically assigned when the service starts.
PowerCenter Integration Service	Dynamic	Dynamically assigned when the service starts.
PowerCenter Repository Service	Dynamic	Dynamically assigned when the service starts.
Reporting and Dashboards Service	Static	HTTP: 10370 HTTPS: No default port number. Manually assigned during service creation.
Reporting Service	Static	HTTP: 16080 HTTPS: No default port number.

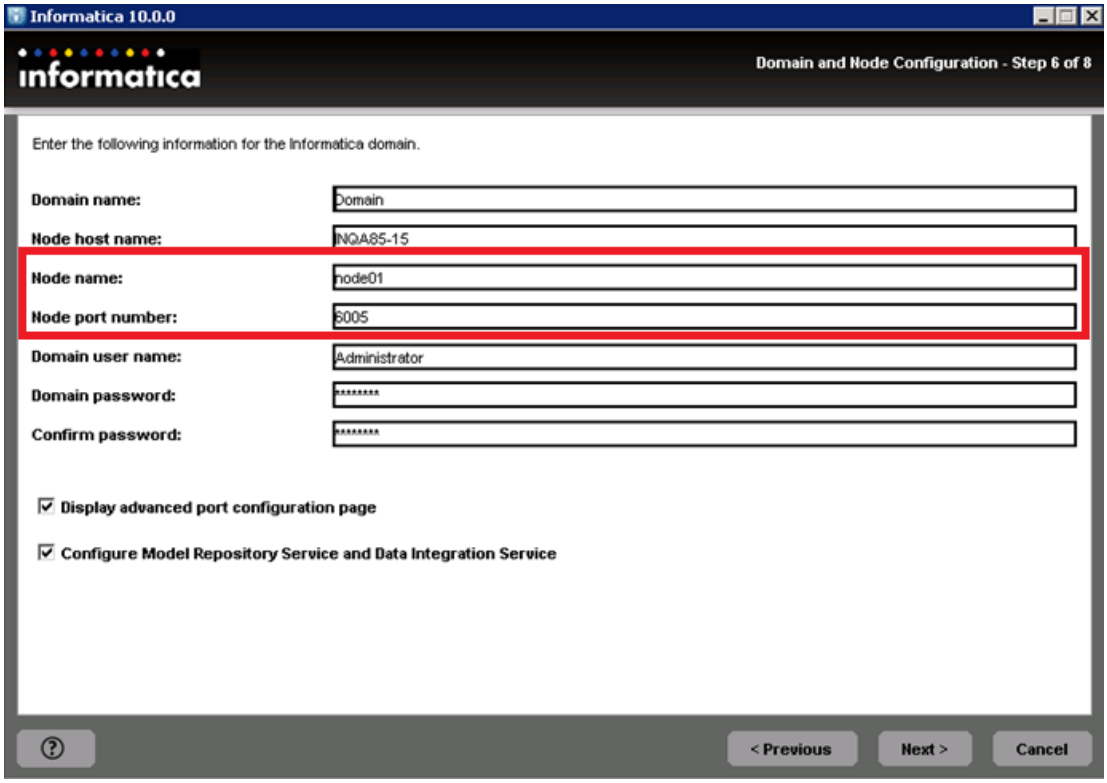
Service Name	Static or Dynamic?	Port Numbers
SAP BW Service	Static	Type R DEST: Refers to an entry in the saprfc.ini SAP connection configuration file. The GWSERV server name in the DEST entry picks a TCP/IP port from the \WINDOWS \system32\drivers\etc\services or \etc\services file.
The Search Service	Static	HTTP: 8084
Web Services Hub	Static	HTTP: 7333 HTTPS: 7343 Internal Port Number: 15555

Static Port Configuration

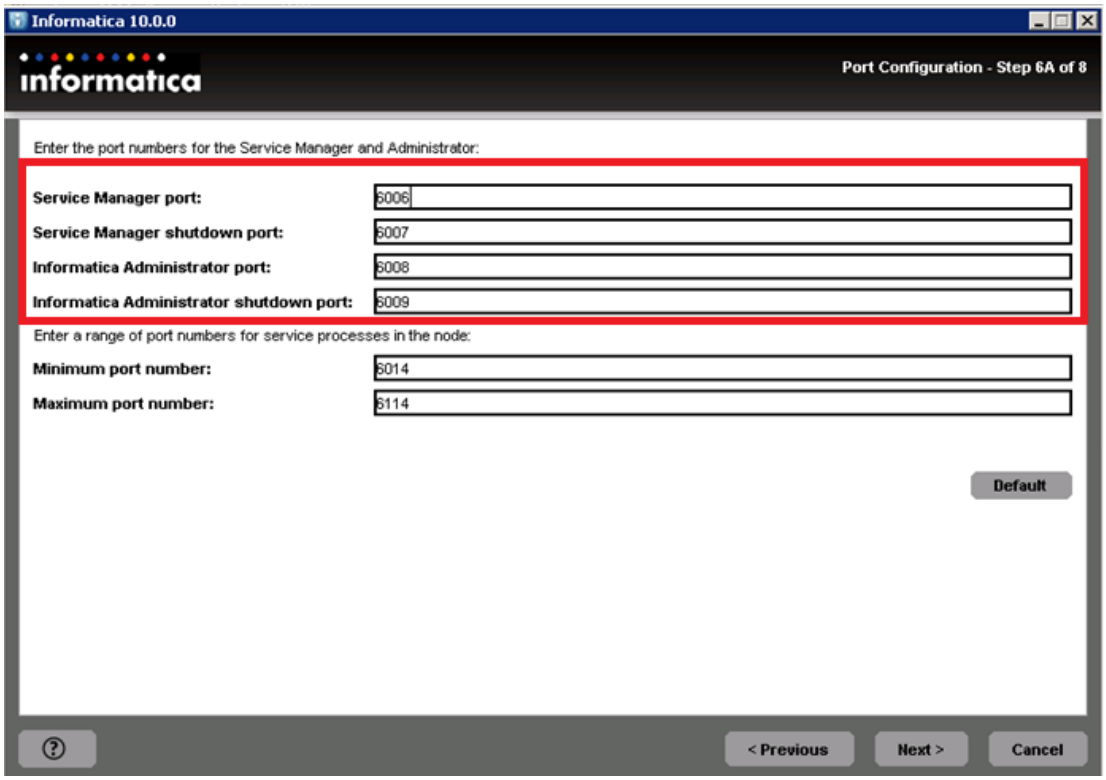
Defining the Port Numbers for the Informatica Domain (Graphical Mode)

The port numbers are defined in the following pages of the Informatica installer.

Domain and Node Configuration



Port Configuration



Use the following guidelines to determine the port numbers:

- The port number you specify for the domain and for each component in the domain must be unique.
- The port number for the domain and domain components cannot be within the range of the port numbers that you specify for the application service processes.
- The highest number in the range of port numbers that you specify for the application service processes must be at least three numbers higher than the lowest port number. For example, if the minimum port number in the range is 6400, the maximum port number must be at least 6403.
- The port numbers that you specify cannot be lower than 1025 or higher than 65535.

Defining the Port Numbers for the Informatica Domain (Command Line)

Run the infasetup DefineDomain command with the following options to define the port numbers.

```
infasetup DefineDomain
<-NodeAddress|-na> node_host:port
[<-ServiceManagerPort|-sp> service_manager_port]
[<-ServerPort|-sv> server_shutdown_port]
[<-AdminconsolePort|-ap> admin_tool_port]
[<-AdminconsoleShutdownPort|-asp> admin_tool_shutdown_port]
[<-HttpsPort|-hs> https_port]
```

Modifying the Port Numbers for the Informatica Domain (Command Line)

Run the infasetup UpdateGatewayNode command with the following options to modify the port numbers.

```
infasetup UpdateGatewayNode
[<-NodeAddress|-na> node_host:port]
[<-ServiceManagerPort|-sp> service_manager_port]
[<-AdminconsolePort|-ap> admin_tool_port]
```

```
[<-HttpsPort|-hs> https_port]
[<-ServerPort|-sv> server_shutdown_port]
[<-AdminconsoleShutdownPort|-asp> admin_tool_shutdown_port]
```

Defining the Static Port Number for an Informatica Service (Informatica Administrator)

From Informatica Administrator, define the port properties when you create the service.

Defining the Static Port Number for an Informatica Service (Command Line)

Analyst Service

When you run the infacmd as createService command to create the Analyst Service, set the following option to define the port.

```
infacmd.sh as createService
<-HttpPort> http_port
```

Content Management Service

When you run the infacmd cms createService command to create the Content Management Service, set the following option to define the port.

```
infasmd.sh cms createService
[<-HttpPort> http_port]]
```

Data Integration Service

When you run the infacmd dis createService command to create the Data Integration Service, set the following options to define the ports.

```
infacmd.sh dis createService
[<-ServiceOptions|-so> option_name=value ...(GeneralOptions.HttpPort, GeneralOptions.HttpsPort)]
```

Metadata Manager Service

When you run the infacmd isp createMMService command to create the Metadata Manager Service, set the following option to define the port.

```
infasmd.sh isp createMMService
[<-ServiceOptions|-so> option_name=value ...(PortNumber, AgentPort)]
```

Reporting Service

When you run the infacmd isp createReportingService command to create the Reporting Service, set the following option to define the port.

```
infacmd.sh isp createReportingService
[<-ServiceOptions|-so> option_name=value ...(HttpPort, sslPort)]
```

The Search Service

When you run the infacmd search createService command to create the Search Service, set the following option to define the port.

```
infacmd.sh search createService
[<-SearchServicePort|-sp>] search_service_port_number]
```

Web Services Hub

When you run the infacmd isp createWSHubService command to create the Web Services Hub, set the following options to define the ports.

```
infacmd.sh isp createWSHubService
[<-ServiceOptions|-so> option_name=value ...(HubPortNumber, httpsHubPortNumber,
InternalPortNumber)]
```

Modifying the Static Port Number for an Informatica Service (Informatica Administrator)

From Informatica Administrator, edit the service properties.

Modifying the Static Port Number for an Informatica Service (Command Line)

Analyst Service

Run the infacmd as updateServiceProcessOptions command with the following options to modify the port number.

```
infasmd.sh as updateServiceProcessOptions
[<-Options|-o> option_name=value ...(GeneralOptions.HttpPort)]
```

Content Management Service

Run the infacmd cms updateServiceProcessOptions command with the following options to modify the port numbers.

```
infasmd.sh cms updateServiceProcessOptions
[<-Options|-o> option_name=value ...(GeneralOptions.HttpPort)]
```

Data Integration Service

Run the infacmd dis updateServiceProcessOptions command with the following options to modify the port number.

```
infacmd.sh dis updateServiceProcessOptions
[<-ServiceOptions|-so> option_name=value ...( GeneralOptions.HttpPort,
GeneralOptions.HttpsPort)]
```

Metadata Manager Service

Run the infacmd isp updateMMService command with the following options to modify the port numbers.

```
infacmd.sh isp updateMMService
[<-ServiceOptions|-so> option_name=value ...(PortNumber, AgentPort)]
```

Reporting Service

Run the infacmd isp updateReportingService command with the following options to modify the port numbers.

```
infacmd.sh isp updateReportingService
[<-ServiceOptions|-so> option_name=value ...(HttpPort, sslPort)]
```

The Search Service

Run the infacmd search updateServiceOptions command with the following option to modify the port number.

```
infacmd.sh search updateServiceOption
[<-Options|-o> option_name=value ...(SearchServicePort)]
```

Web Services Hub

Run the infacmd isp updateWSHubService command with the following options to modify the port numbers.

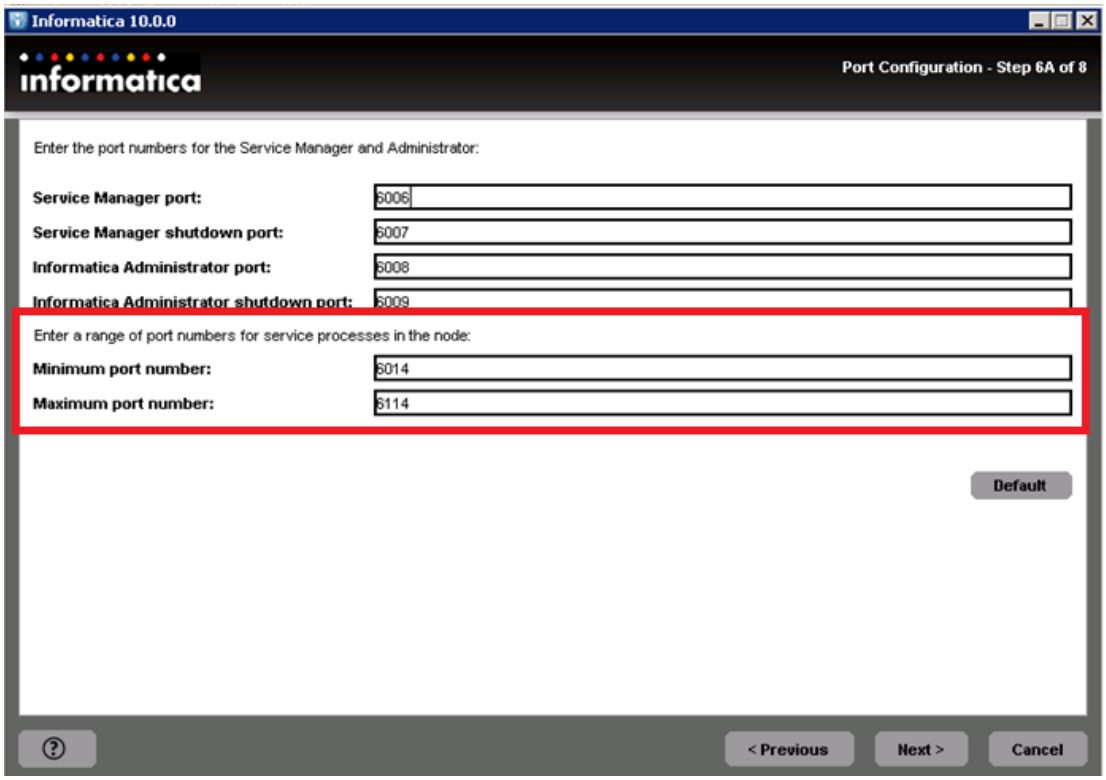
```
infacmd.sh isp updateWSHubService
[<-ServiceOptions|-so> option_name=value ...(HubPortNumber, httpsHubPortNumber,
InternalPortNumber)]
```

Dynamic Port Configuration

The port range represents the range of port numbers that the Informatica Service Manager uses to dynamically assign port numbers to the application service processes.

Defining the Range of Port Numbers for the Informatica Services (Graphical Mode)

If you are installing Informatica in graphical mode, the range of port numbers is specified in the Port Configuration page of the Informatica installer.



When you specify the port range during the installation, at a minimum, the number of ports in the range must be at least twice the number of application service processes that will run on the node.

Defining the Range of Port Numbers for the Informatica Services (Command Line)

If you are installing Informatica using the command line, run the infasetup DefineDomain command with the following options to define the range of port numbers.

```
infasetup defineDomain
<-MinProcessPort|-mi> minimum_port
<-MaxProcessPort|-ma> maximum_port
```

When you specify the port range during the installation, at a minimum, the number of ports in the range must be at least twice the number of application service processes that will run on the node.

Modifying the Range of Port Numbers for the Informatica Services (Informatica Administrator)

From Informatica Administrator, define the range of port numbers through the node Properties view.

Modifying the Range of Port Numbers for the Informatica Services (Command Line)

Run the infacmd isp updateNodeOptions command with the following options to modify the range of port numbers.

```
infasmd.sh isp updateNodeOptions
[<-NodeOptions|-no> option_name=value ...(MaxProcessPort, MinProcessPort]
```

Author

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